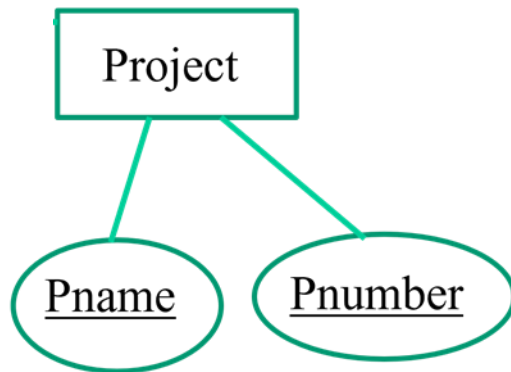


[FIX] Step 1: Strong Entities

- NOTE: in the E/R diagram for translation both Pname and Pnumber are labelled to be key attributes. Therefore Pnumber should be underlined to reflect this in the corresponding diagram.



Employee

<u>SSN</u>	Fname	Lname	Birdate
------------	-------	-------	---------

Department

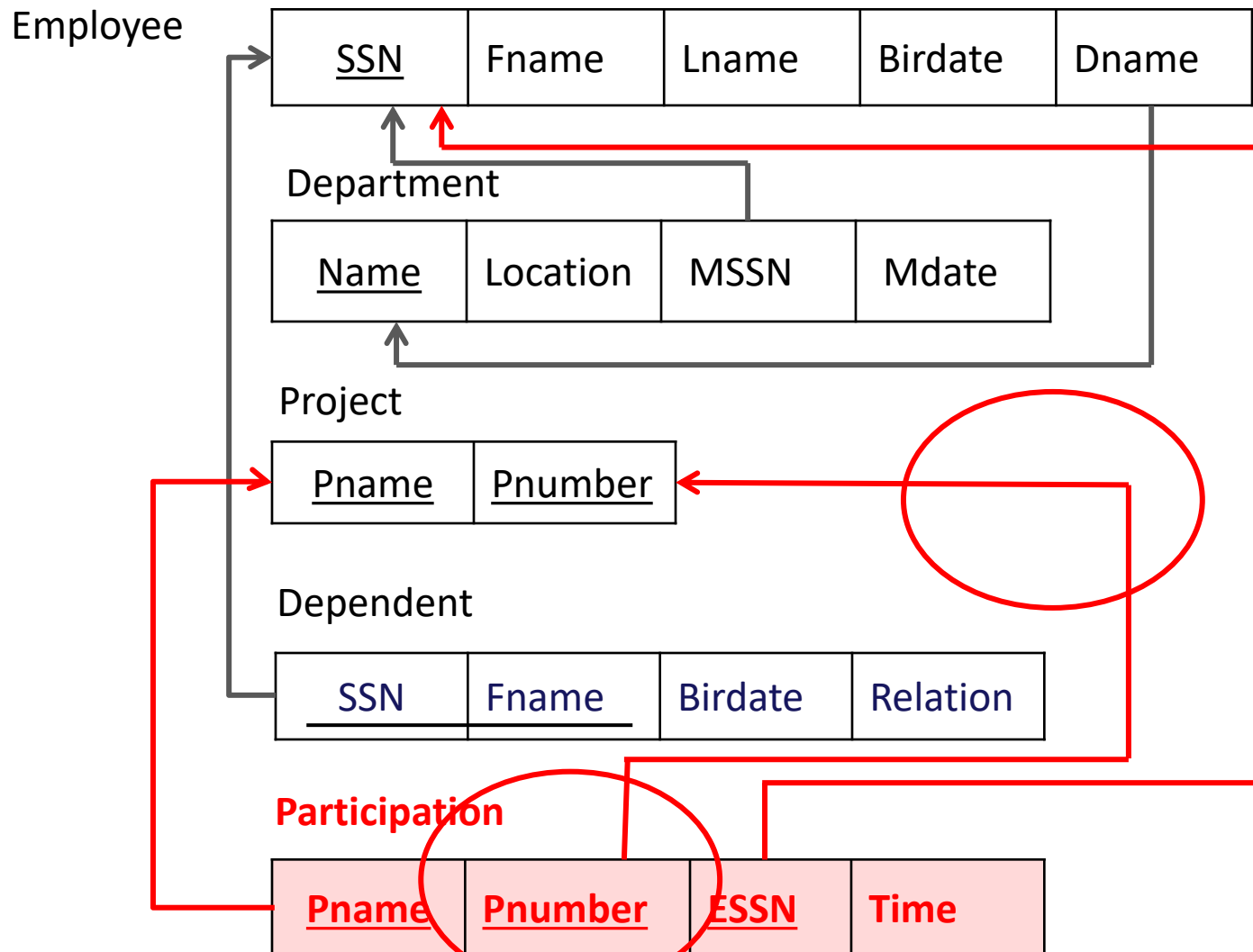
<u>Name</u>	Location
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Project

<u>Pname</u>	<u>Pnumber</u>
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[FIX] Step 5 - M:N Relationships

- NOTE: Pnumber is therefore also included in all roles as foreign key



CSCI3170

Introduction to Database Systems

3-relational_algebra

Slides: Michael Yu

Basic Operators

- Query languages specialized languages for asking questions or queries, that involve data in database
 - Relational algebra is a procedural query language, defining a step-by-step procedure for computing a desired answer
- Queries in algebra are composed of a collection of operators
 - Operators take one or two relations as inputs and produce a new relation as a result.
 - Standard operations produce anonymous (unnamed) temporary relations.
 - Six basic/fundamental operators :
 - select (σ)
 - project (Π)
 - union ($\cup$)
 - set different ($-$)
 - Cartesian product ($\times$)
 - rename (ρ)

Select Operation

- In practice, selection operation σ specifies the tuples to retain through a selection predicate P (a Boolean expression)
 - Notation: $\sigma_P(r)$
 - $\sigma_P(r) = \{t \mid t \in r \wedge P(t)\}$
- P can involve multiple conditions/terms using logical connectives
 - term is :
 $\langle \text{attribute1} \rangle \text{ op } \langle \text{attribute2} \rangle$
OR $\langle \text{attribute} \rangle \text{ op } \langle \text{constant} \rangle$
(op: =, $\neq$, >, $\geq$, <, $\leq$)
 - P consisting of terms connected by $\wedge$ (and), $\vee$ (or), $\neg$ (not)

Relation R

a	b	c	d
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$\sigma_{a=b \wedge d > 5}(R)$

a	b	c	d
α	α	1	7
β	β	23	10

ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

$\sigma_{\text{dept_name} = \text{"Phyics"}}(\text{instructor})$

Project Operation

- Notation: $\Pi_{A1,A2,\dots,Ak}(r)$
 - $A1, A2, \dots, Ak$ as attribute names
 - r is name of relation
- Π allows us to extract columns from a relation
 - subscript specifies fields to be retained (other fields are ‘projected out’)
 - **Relations obey set semantics**
 - Duplicated rows are eliminated in the final result, this follows from definition.

Relation R

a	b	c
α	10	1
α	20	1
β	30	1
β	40	2

 $\xrightarrow{\Pi_{a,c}(R)}$

a	c
α	1
β	1
β	2

ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
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83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

$\Pi_{ID,name,salary}(instructor)$

Practice: Q1

- Find all employees who works in department 4 and whose salary is greater than 25000?

Employee

f_name	l_name	<u>id</u>	sex	salary	superid	dno
Joseph	Chan	999999	M	29500	654321	4
Victor	Wong	001100	M	30000	888555	5
Carrie	Kwan	898989	F	26000	654321	4
Joyce	Fong	345345	F	12000	777888	4

$$\sigma_P(r)$$
$$\Pi_{A_1, A_2, \dots, A_k}(r)$$

Practice: Q2

- Find the employee names and department number of all employees

Employee

f_name	l_name	<u>id</u>	sex	salary	superid	dno
Joseph	Chan	999999	M	29500	654321	4
Victor	Wong	001100	M	30000	888555	5
Carrie	Kwan	898989	F	26000	654321	4
Joyce	Fong	345345	F	12000	777888	4

$$\sigma_P(r)$$
$$\Pi_{A_1, A_2, \dots, A_k}(r)$$

Practice: Q3

- Find the sex and department number of all employees

Employee

f_name	l_name	<u>id</u>	sex	salary	superid	dno
Joseph	Chan	999999	M	29500	654321	4
Victor	Wong	001100	M	30000	888555	5
Carrie	Kwan	898989	F	26000	654321	4
Joyce	Fong	345345	F	12000	777888	4

$$\sigma_P(r)$$
$$\Pi_{A_1, A_2, \dots, A_k}(r)$$

Practice: Q4

- Find employee names and department number of all employees who works in department 4 and whose salary is greater than 25000?

Employee

f_name	l_name	id	sex	salary	superid	dno
Joseph	Chan	999999	M	29500	654321	4
Victor	Wong	001100	M	30000	888555	5
Carrie	Kwan	898989	F	26000	654321	4
Joyce	Fong	345345	F	12000	777888	4

- Recall: Result of a relational-algebra expr. is always a relation.

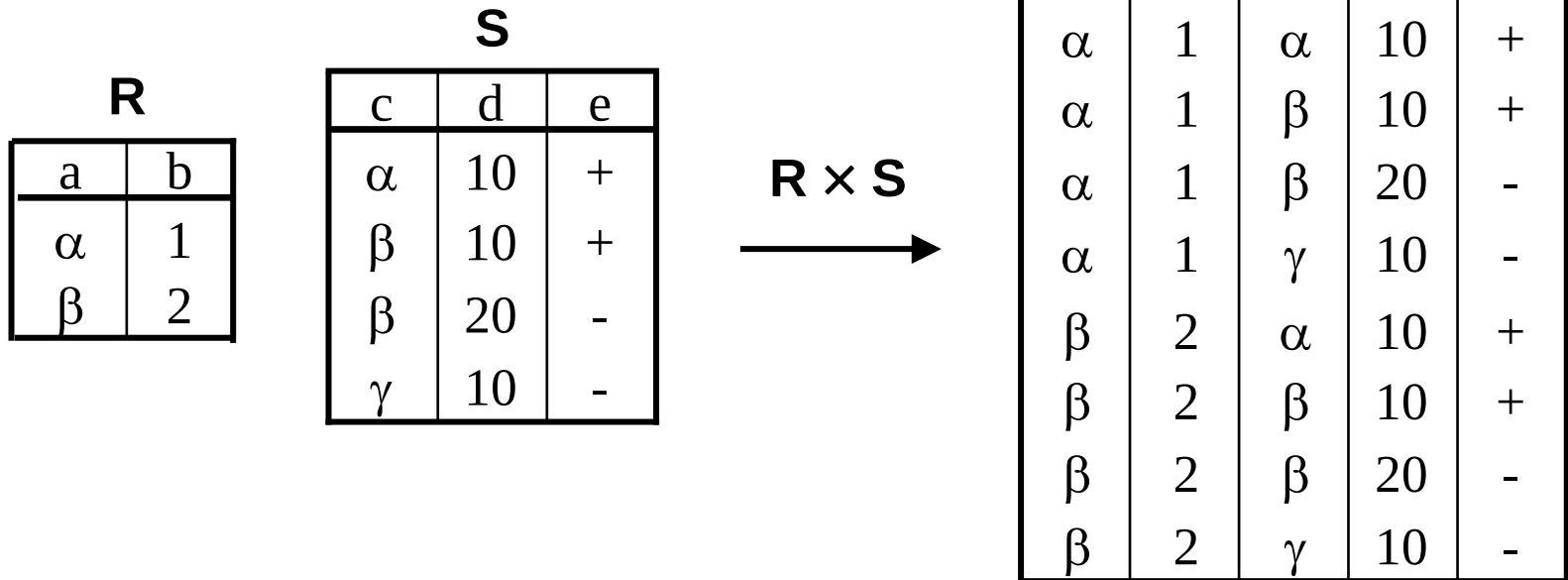
$$\sigma_P(r)$$
$$\Pi_{A_1, A_2, \dots, A_k}(r)$$

Solutions

- $\sigma_{\text{dno}=4 \wedge \text{salary}>25000} (\text{Employee})$
- $\Pi_{\text{f_name}, \text{l_name}, \text{dno}} (\text{Employee})$
- $\Pi_{\text{sex}, \text{dno}} (\text{Employee})$
- $\Pi_{\text{f_name}, \text{l_name}} (\sigma_{\text{dno}=4 \wedge \text{salary}>25000} (\text{Employee}))$

Cartesian-Product

- Notation for reference: $r \times s = \{pq \mid p \in r \wedge q \in s\}$
 - Cartesian product (Cross product)
 - returns a relation instance whose schema contains all the fields of R followed by all the fields of S
 - result contains one tuple $\langle r, s \rangle$ (concatenation of tuples r and s) for each pair of tuples $r \in R, s \in S$



Guiding Demonstration (1)

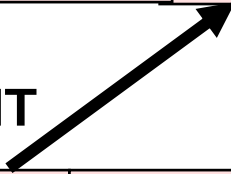
- Retrieve for each female employee, a list of the names of her dependents.

EMPLOYEE

f_name	l_name	<u>id</u>	bdate	addr	sex	salary	super_id	dno
--------	--------	-----------	-------	------	-----	--------	----------	-----

DEPENDENT

<u>eid</u>	<u>dependent-name</u>	sex	bdate	relationship
------------	-----------------------	-----	-------	--------------



Guiding Demonstration (2)

EMPLOYEE

f_name	l_name	id	bdate	addr	sex	salary	super_id	dno
--------	--------	----	-------	------	-----	--------	----------	-----

DEPENDENT

<u>eid</u>	<u>dependent-name</u>	sex	bdate	relationship
------------	-----------------------	-----	-------	--------------



$\text{Female_emps} \leftarrow \sigma_{\text{sex}='F'} (\text{Employee})$

$\text{Empnames} \leftarrow \Pi_{\text{f_name}, \text{l_name}, \text{id}} (\text{Female_emps})$

Female_emps

f_name	l_name	id	bdate	address	sex	salary	superid	dno
Alicia	Chan	998877	2-Jul-70	231, Cai Road, HK	F	9500	654321	4
Jennifer	Wong	654321	20-June-60	342, Cheung Road, HK	F	30000	888555	4
Joyce	Fong	345345	19-Dec-80	23, Young Road, HK	F	12000	777888	5

Empnames

f_name	l_name	id
Alicia	Chan	998877
Jennifer	Wong	654321
Joyce	Fong	345345

Guiding Demonstration (3)

EMPLOYEE

f_name	l_name	<u>id</u>	bdate	addr	sex	salary	super_id	dno
--------	--------	-----------	-------	------	-----	--------	----------	-----

DEPENDENT

<u>eid</u>	<u>dependent-name</u>	sex	bdate	relationship
------------	-----------------------	-----	-------	--------------

Empnames

f_name	l_name	id
Alicia	Chan	998877
Jennifer	Wong	654321
Joyce	Fong	345345

Dependents

eid	dep_name	sex	bdate	relationship
334455	Alice	F	5-Apr-90	Daughter
334455	Theodore	M	3-Mar-92	Son
654321	Abner	M	29-Feb-94	Son
123456	Alice	F	2-Nov-97	Daughter

Empnames

f_name	l_name	id
Alicia	Chan	998877
Jennifer	Wong	654321
Joyce	Fong	345345

Dependents

eid	dep_name	sex	bdate	relationship
334455	Alice	F	5-Apr-90	Daughter
334455	Theodore	M	3-Mar-92	Son
654321	Abner	M	29-Feb-94	Son
123456	Alice	F	2-Nov-97	Daughter

Guiding Demonstration (4)

Emp_dependents ← Empnames × Dependents

Emp_dependents

f_name	l_name	id	eid	dep_name	sex	bdate	relationship
Alicia	Chan	998877	334455	Alice	F	5-Apr-90	Daughter
Alicia	Chan	998877	334455	Theodore	M	3-Mar-92	Son
Alicia	Chan	998877	654321	Abner	M	29-Feb-94	Son
Alicia	Chan	998877	123456	Alice	F	2-Nov-97	Daughter
Jennifer	Wong	654321	334455	Alice	F	5-Apr-90	Daughter
Jennifer	Wong	654321	334455	Theodore	M	3-Mar-92	Son
Jennifer	Wong	654321	654321	Abner	M	29-Feb-94	Son
Jennifer	Wong	654321	123456	Alice	F	2-Nov-97	Daughter
Joyce	Fong	345345	334455	Alice	F	5-Apr-90	Daughter
Joyce	Fong	345345	334455	Theodore	M	3-Mar-92	Son
Joyce	Fong	345345	654321	Abner	M	29-Feb-94	Son
Joyce	Fong	345345	123456	Alice	F	2-Nov-97	Daughter

Guiding Demonstration (5)

Actual_dependents $\leftarrow \sigma_{id=eid} (\text{Emps_dependents})$

Actual_dependents

f_name	l_name	id	eid	dep_name	sex	bdate	relationship
Jennifer	Wong	654321	654321	Abner	M	29-Feb-94	Son

Result $\leftarrow \Pi_{f_name, l_name, dep_name} (\text{Actual_dependents})$

Result

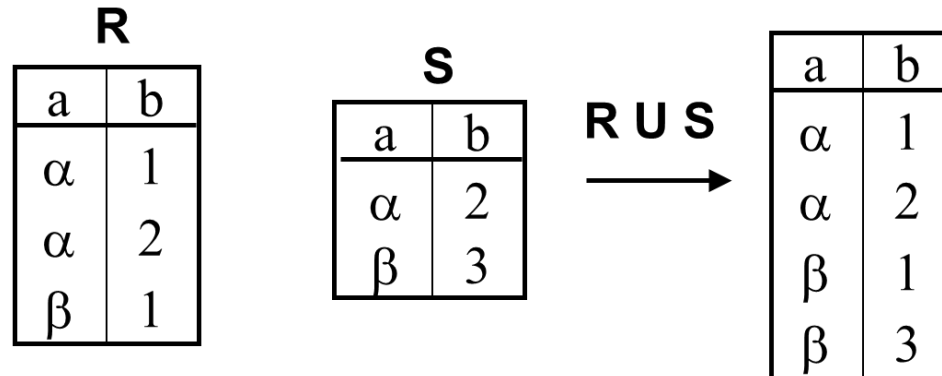
f_name	l_name	dep_name
Jennifer	Wong	Abner

No more than a **convenience**

- Note the ($\leftarrow$) symbol from past symbols
 - Known as the “assignment operator”
 - Write query as a sequential program consisting of
 - a series of assignments
 - result of the expression right of ($\leftarrow$) assigned to a temporary **relation variable** left of ($\leftarrow$)
 - temporary relation variable can be referenced in later steps
- Purely a notational and procedural **convenience** added later in database textbooks
 - **NOT** one of Edgar F. Codd's original pure mathematical operators
 - formally, relational algebra outputs are unnamed by default, except when you explicitly use the rename operator (ρ)

Self-Reading: Union (1)

- Notation: $r \cup s$
 - $r \cup s = \{t \mid t \in r \vee t \in s\}$
 - commutative: $R \cup S = S \cup R$
 - schema of the result is defined to be identical to the schema of R
 - fields of $R \cup S$ inherit names from R
 - returns a relation instance containing all tuples that occur in either relation R or relation instance S (or both)



- Example:

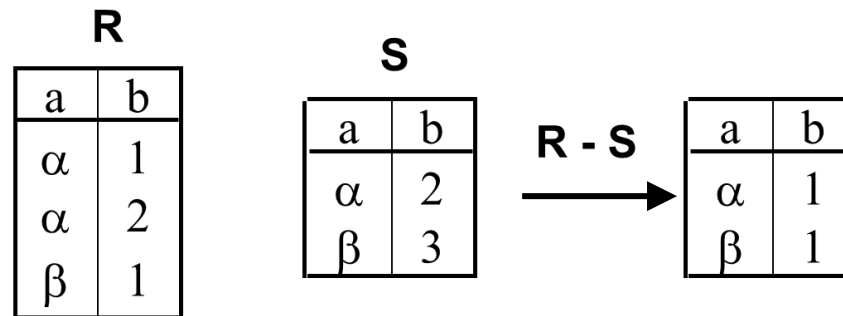
$$\Pi_{course_id} (\sigma_{semester="Fall" \wedge year=2009}(section)) \cup \Pi_{course_id} (\sigma_{semester="Spring" \wedge year=2010}(section))$$

Important Note on Compatability

- Intuitively, for $R \cup S$ to be valid, R and S ...
 - Must have *same* **arity** (same number of attributes)
 - Corresponding fields have same domains
- This requirement generally known as *union-compatibly*, the same requirements apply to set-difference, and set-intersection as well

Self-Reading: Set Difference

- Notation: $r - s$
 - $r - s = \{t \mid t \in r \wedge t \notin s\}$
 - taken between **union-compatible** relations
 - $R - S$ returns a relation instance containing all the tuples that occur in R but not in S
 - **not commutative**: $R - S \neq S - R$.
 - Schema of the result is defined to be identical to the schema of R



- Example:

$$\Pi_{course_id} (\sigma_{semester="Fall" \wedge year=2009} (section)) - \Pi_{course_id} (\sigma_{semester="Spring" \wedge year=2010} (section))$$

Rename Operation

- $\rho_X(E)$ returns the expression E under the name X
 - Allows us to name, and therefore to refer to, the results of relational algebra expressions (Allow us to refer to a relation by more than one name)
- If a relational-algebra expression E has arity n , then $\rho_{X(A_1, A_2, \dots, A_n)}(E)$ returns result of expression E under the name X with **attributes renamed** to $A_1, A_2, \dots, A_n$
- *Why is Rename a basic operator where assignment is not usually considered as one?*

Example **

- Find **largest** salary in the university
 - Step 1:** find instructor salaries that are less than some other instructor salary (i.e. not maximum)

$$\Pi_{instructor.salary} (\sigma_{instructor.salary < d.salary} (instructor \times \rho_d (instructor)))$$

- Step 2:** Find largest salary

$$\Pi_{salary} (instructor) - \Pi_{instructor.salary} (\sigma_{instructor.salary < d.salary} (instructor \times \rho_d (instructor)))$$

instructor

ID	name	dept_name	salary
8	ABC	SEEM	100
7	XYZ	SEEM	120

I.ID	I.name	I.dept_name	I.salary	d.ID	d.name	d.salary	d.salary
8	ABC	SEEM	100	8	ABC	SEEM	100
8	ABC	SEEM	100	7	XYZ	SEEM	120
7	XYZ	SEEM	120	8	ABC	SEEM	100
7	XYZ	SEEM	120	7	XYZ	SEEM	120

Cartesian-Product Operation

- Notation: $w(W)$
 - relation w following schema W
- In most operations, e.g., cartesian-products, there is an assumption that attributes of $r(R)$ and $s(S)$ are disjoint. (That is, $R \cap S = \emptyset$)
- So... if attributes of $r(R)$ and $s(S)$ were not disjoint, then some renaming must be used (and is).
 - Since this does happen a lot, naming schema devised to distinguish between the attribute names if they are the same in $r(A, B)$ and $s(A, C)$
 - Simply attach the relation name, $r.A$ and $s.A$ (known as **dot-notation**)

Reference

- Some slides adapted from Database System Concepts – 6th Edition by Silberschatz, Korth and Sudarshan